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ABSTRACT

Multiple quantum well (MQW) materials composed of $Pb_{0.81}Sn_{0.19}Se$ wells and $Pb_{0.80}Sr_{0.20}Se$ barriers with intersubband energy gaps of 343 meV and 450 meV were modeled for thermophotovoltaic (TPV) device performance. The effect of L-valley degeneracy removal in these (111)-oriented IV-VI semiconductor quantum wells was evaluated. Degeneracy splitting reduces the effective densities of states in both the valence and conduction bands. Thermally generated intrinsic charge carrier concentrations are smaller by a factor of three as compared to bulk materials with the same bandgap energies. A current-matched dual junction TPV cell made from these MQW materials in a generator with a 1215°C radiator is predicted to have a power density of 2.34 W/cm^2, 49% better than the power density generated by a cell made from bulk materials with the same bandgap energies.

INTRODUCTION

Thermophotovoltaic (TPV) conversion of mid-infrared thermal radiation to electrical power offers many advantages over other methods of hydrocarbon combustion based electrical power generation. Solid state TPV devices are small, reliable, and quiet. These features facilitate development of highly efficient combined heat and power (CHP) systems. For example, water and air heating systems can be equipped with TPV cells placed adjacent to a combustion heated radiator. The combustion heat not converted to electricity by the TPV cell is then used to heat water or air. Fuel utilization efficiency associated with such TPV/CHP systems can be better than 95% based on fuel utilization efficiencies for modern gas-fired water heaters and furnaces. By contrast, when electrical power is generated by centralized power plants, fuel utilization efficiencies are only 33% due to combustion heat and transmission line losses. The economic benefit of a more than 60% lower fuel cost for CHP electricity can drive commercial adoption of a practical TPV/CHP technology. Since highly fuel efficient TPV/CHP generation replaces less efficient central power generation, a clear outcome of this adoption will be a reduction in CO_2 emissions. However, TPV adoption has been hindered by the high cost of currently available TPV cells. This paper describes a novel semiconductor materials system that can significantly reduce the cost of TPV cells. In addition, these new materials will enable fabrication of TPV cells with four times the power generation density of currently available TPV cells.

Narrow bandgap semiconductors are required for photovoltaic conversion of mid-infrared thermal radiation to electrical power. III-V semiconductors such as binary GaSb (E_g = 720 meV) and quaternary GaInAsSb (E_g = 530 meV) grown on GaSb substrates have been used to fabricate TPV cells. TPV devices based on these materials systems exhibit electrical power densities of 0.6 W/cm^2 with a 1215°C thermal radiator.[1] GaSb-based TPV cells cost more than \$60/watt, where more than 70% of this cost is due to expensive GaSb substrate material. TPV cell fabrication cost can be decreased significantly with the use of thin layers of narrow bandgap materials grown on a low cost substrate such as silicon. IV-VI semiconductors offer such a solution. IV-VI semiconductors, i.e. PbSe and related alloys, are soft materials with an easy dislocation glide system in the <111> directions along the {100} planes. This allows plastic deformation without crystalline defect formation when layers are grown on (111)-oriented substrates.[2] III-V semiconductor materials lack this feature, which has complicated development of a reliable III-V-on-Si materials technology.[3] IV-VI-on-Si heteroepitaxy has been demonstrated[4–6] where high epilayer crystalline quality is maintained even after repeated thermal cycling.[7]

Although IV-VI semiconductors are attractive for fabricating low cost TPV cells, the electronic band structure of bulk IV-VI materials is not favorable for TPV device operation. IV-VI semiconductors have direct bandgaps at the L-point in **k**-space. Fourfold degeneracy at the L-point results in high densities of states at the band extrema and high concentrations of thermally generated charge carriers, resulting in high reverse bias saturation currents and reduced open circuit voltages. However, growth of multiple quantum well (MQW) materials on (111)-oriented substrates creates quantum confinement of electrons and holes in the [111] direction and removes L-valley degeneracy.[8] This paper describes our analysis of PbSnSe/PbSrSe MQW materials. It demonstrates a novel use of MQW materials to improve photovoltaic device performance. It is the first known report of how open circuit voltage can be increased by removing band degeneracy to reduce the densities of states at the bandgap.

MQW MATERIALS DESIGNS

Table I lists band structure parameters for four different IV-VI semiconductor alloy compositions. The first two alloys are $Pb_{0.937}Sr_{0.063}Se$ (E_g=450 meV) and $Pb_{0.976}Sr_{0.024}Se$ (E_g=343 meV).[9] These alloys have been grown by molecular beam epitaxy (MBE) on PbSe, BaF_2, and Si substrates with high epilayer quality.[10] Both n-type and p-type layers can be obtained with controllable carrier concentrations between about $1x10^{17}$ cm^{-3} and $1x10^{19}$ cm^{-3} and mobilities between about 300 cm^2/Vs and 600 cm^2/Vs. If minority carrier lifetimes are in the range of 10 ns, then charge carrier diffusion lengths are longer than 3 μm, which is a good match for the thickness of MBE-grown pn junction structures. Mid-IR lasers have been fabricated using PbS/PbSrS heterojunctions grown by MBE on Si, and laser threshold measurements showed a minority carrier lifetime of 10 ns.[11] These experimental data allow a realistic model for TPV device performance to be developed.

TABLE I. Band structure parameters for IV-VI semiconductor alloys at 300 K.[9,13]

Alloy (T=300K)	E_g (meV)	Conduction Band m_l	Conduction Band m_t	Valence Band m_l	Valence Band m_t
$Pb_{0.937}Sr_{0.063}Se$	450	0.140	0.070	0.140	0.070
$Pb_{0.976}Sr_{0.024}Se$	343	0.108	0.054	0.108	0.054
$Pb_{0.80}Sr_{0.20}Se$	840	0.257	0.129	0.257	0.129
$Pb_{0.81}Sn_{0.19}Se$	84	0.044	0.025	0.044	0.023

The bandgaps of PbSrSe alloys are smaller than those that can be reliably obtained with III-V semiconductor alloys. This significantly increases the amount of mid-IR emission that can be absorbed from a thermal radiator. For example, a 1215°C thermal radiator with an emissivity of 0.9 emits 4.20 W/cm^2 of optical power up to the 1.7 μm cut-off wavelength for GaSb, whereas an additional 12.7 W/cm^2 of optical power is available up to the 3.6 μm cut-off wavelength for $Pb_{0.976}Sr_{0.024}Se$. This radiation will also be readily absorbed by thin MBE-grown films because above-bandgap absorption coefficients for PbSe and related materials are in the range of $1x10^4$ cm^{-1}.[12] For example, a 4 μm thick layer of PbSe will absorb more than 98% of non-reflected incident radiation according to Beer's law.

The other two alloys listed in Table I are $Pb_{0.80}Sr_{0.20}Se$ (E_g = 840 meV)[9] and $Pb_{0.81}Sn_{0.19}Se$ (E_g = 84 meV).[13] These are the compositions of the barrier and well layers, respectively, for growth of MQW materials that can be designed to have subband gaps that match the bandgaps of the first two alloys. An upper limit of 20% Sr in the PbSrSe alloy exists for a practical TPV device material due to doping difficulty and reduced charge carrier mobility. On the other hand, MBE-grown PbSnSe alloys are easily doped and have high charge carrier mobilities.[14] For example, unpublished Hall effect data for a Bi-doped PbSnSe layer (3.3 μm thick) grown with a 15% SnSe/PbSe flux ratio showed a room temperature bandgap of 128 meV, an electron concentration of 3 x 10^{18} cm^{-3}, and an electron mobility of 1,588 cm^2/Vs. Experimental data demonstrating the ability to grow such high quality narrow bandgap material on silicon[15] provides strong validation of the modeling work described here. The modeling work begins with using known IV-VI semiconductor band structure parameters to calculate effective densities of states for bulk and MQW materials. Measured materials parameter (doping, mobility, lifetime) are then used to calculate intrinsic carrier density and reverse bias saturation current under dark testing conditions, both of which are decreased when lower density of states MQW materials are used. Improvements in the open circuit voltages are obtained as a result.

Band structure parameters for electrons and holes in IV-VI semiconductors are given in terms of transverse and longitudinal masses that define ellipsoids of constant energy in **k**-space. The longitudinal axes of L-valley ellipsoids lie along the eight different {111} directions in three-dimensional **k**-space. Effective masses for electrons and holes in isotropic bulk materials are calculated using $m = \left(m_l m_t^2\right)^{\frac{1}{3}}$. Charge carrier effective masses in anisotropic MQW materials, however, must take into account the L-valley band structure. There are four L-valleys per unit cell (Brillouin zone) in **k**-space. A MQW material grown on a (111)-oriented substrate will have quantum confinement in the [111] direction, and only one of the four L-valleys will have its longitudinal axis aligned in this direction, which is normal to the (111) plane in **k**-space. The mass for charge carriers in this normal L-valley will be $m_{normal} = m_o m_l$, while the mass for charge carriers in the other three L-valleys that are at oblique angles with respect to the [111] direction is given by $m_{oblique} = m_o\left[\frac{1}{9}\left(\frac{8}{m_t} + \frac{1}{m_l}\right)\right]^{-1}$.[16] In both cases m_o is the free electron mass.

Table II lists effective masses for bulk and MQW materials with (111)-oriented $Pb_{0.81}Sn_{0.19}Se$ wells calculated using the above equations and the band structure parameters from Table I.

TABLE II. Electron and hole effective masses in IV-VI semiconductor materials at 300 K.

	Conduction Band			Valence Band		
Material	m_c	m_{normal}	$m_{oblique}$	m_v	m_{normal}	$m_{oblique}$
$Pb_{0.937}Sr_{0.063}Se$	0.088			0.088		
$Pb_{0.976}Sr_{0.024}Se$	0.068			0.068		
$Pb_{0.81}Sn_{0.19}Se$ QWs		0.044	0.026		0.044	0.024

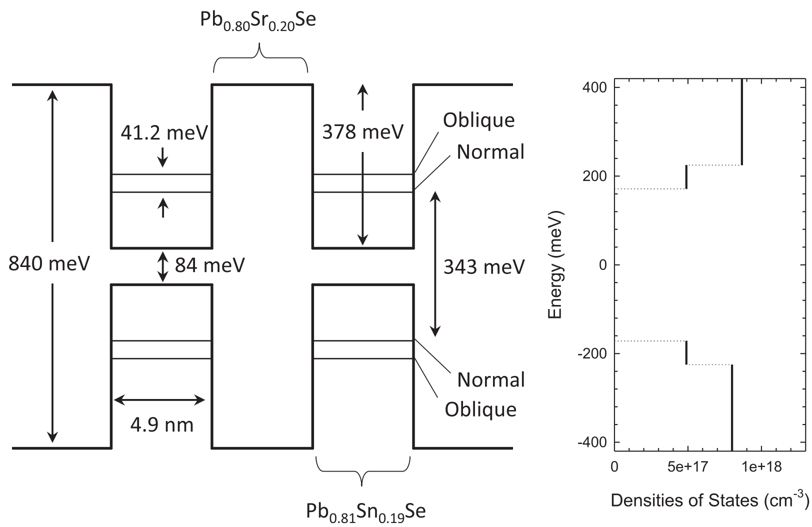

FIG. 1. MQW material with 4.9 nm thick wells (14 monolayers of $Pb_{0.81}Sn_{0.19}Se$ in the [111] direction). Oblique and normal valley subband energies and corresponding subband densities of states are shown.

The symmetric band structure of IV-VI semiconductors is reflected in the same effective mass values for electrons and holes in the $Pb_{0.937}Sr_{0.063}Se$ and $Pb_{0.976}Sr_{0.024}Se$ bulk alloys where the L-valleys are four-fold degenerate. Due to geometric quantum size effects, this L-valley degeneracy is removed in (111)-oriented $Pb_{0.81}Sn_{0.19}Se$ wells. Two different effective masses for electrons and holes in the normal L-valley and the three-fold degenerate oblique L-valleys are obtained.

Schrödinger's equation was used to calculate subband energy levels in quantum wells with depths of 378 meV in both the conduction and valence bands using the effective masses in Table II. Intersubband energy gaps of 450 meV and 343 meV between the lower energy normal valley subbands were obtained with quantum well widths of 3.5 nm (10 monolayers) and 4.9 nm (14 monolayers), respectively. Figure 1 shows an energy level diagram for a MQW material with 4.9 nm thick wells and equal barrier widths. Also shown in Figure 1 is a plot of MQW densities of states, $N_{MQW} = M\frac{mkT}{\pi d\hbar^2}$, where d is MQW periodicity and M is the L-valley degeneracy, which is one for the normal valley subband and three for the oblique valley subbands. A key observation is the lifting of L-valley degeneracy with oblique and normal subband separations of 43.7 meV and 41.2 meV for the 3.5 nm and 4.9 nm MQW materials, respectively.

The higher activation energies for charge carrier population of the oblique valley subbands, 537.4 meV for the 3.5 nm MQW and 425.4 meV for the 4.9 nm MQW, result in significantly smaller concentrations of thermally generated charge carriers in a MQW material as compared to a bulk material with the same bandgap energy. Intrinsic carrier concentrations for bulk and MQW materials, $n_i = \sqrt{N_c N_v}e^{\frac{-E_g}{2kT}}$, are shown in Table III. E_g is the interband optical transition energy, and N_c and N_v are the densities of states in the conduction and valence bands, respectively. The density of states for bulk IV-VI semiconductor materials is $N_{Bulk} = 2M\left(\frac{2\pi mkT}{h^2}\right)^{\frac{3}{2}}$, where M equals 4, the number of degenerate L-valleys. Intrinsic carrier densities for MQW materials are shown for the normal and oblique valley subbands as well as the total, $n_i = n_i{}^n + n_i{}^o$. Note that MQW intrinsic carrier concentrations are about a factor of three smaller than the intrinsic carrier densities for the same bandgap bulk material.

TABLE III. Intrinsic carrier concentrations for bulk and MQW IV-VI semiconductor materials at 300 K.

		Bulk	MQW		
Material	E_g	n_i	$n_i{}^n$	$n_i{}^o$	n_i
$Pb_{0.937}Sr_{0.063}Se$	450	4.62e14			
$Pb_{0.976}Sr_{0.024}Se$	343	2.46e15			
MQW (3.5 nm)	450		1.19e14	3.79e13	1.57e14
MQW (4.9 nm)	343		6.68e14	2.33e14	9.01e14

TPV DEVICE MODEL

High electrical power density is a key TPV cell performance metric. Increased cell power provides the same economic benefit as reduced manufacturing cost on a dollar-per-watt basis. Cell power is the product of short circuit current, J_{sc}, open circuit voltage, V_{oc}, and fill factor, FF. For example, InGaAsSb TPV devices with a 530 meV bandgap energy generated 0.59 W/cm^2 of power where J_{sc}, was 2.9 A/cm^2, V_{oc} was 306 mV, and FF was 0.67.[17] J_{sc}, and V_{oc} are related through $V_{oc} = \frac{kT}{q} ln\left(\frac{J_{sc}}{J_o} + 1\right)$. As discussed above, J_{sc} can be estimated for given combinations of radiant power, bandgap, and EQE. J_o, often referred to as dark current, is the reverse bias saturation current, $J_o = n_i^2\sqrt{\frac{qkT\mu}{\tau}}\left(\frac{1}{n_n} + \frac{1}{p_p}\right)$, under dark testing conditons. It depends strongly on intrinsic carrier concentration as well as charge carrier mobility, pn junction doping levels, and minority carrier lifetimes. Due to the symmetric band structure of IV-VI semiconductors, electron and hole mobilities can be considered equal along with the electron and hole minority carrier lifetimes. These equations, along with known doping levels and charge carrier mobilities, enable determination of V_{oc}.

Figure 2 shows calculated V_{oc} values as a function of J_{sc} for MQW (E_g=343 meV and 450 meV) and bulk (E_g=343 meV and 450 meV) materials. TPV device parameters at room temperature are: n_n = 3x10^{18} cm^{-3}, p_p = 3x10^{17} cm^{-3}, $\mu_n = \mu_p = \mu$ = 350 cm^2/Vs, and $\tau_n = \tau_p = \tau$ = 10 ns. All of these assumed parameters are supported by experimental data for IV-VI semiconductor layers grown by MBE on silicon substrates.[4–10] Figure 3 shows V_{oc} data plotted as a function of minority carrier lifetime. These data simulate TPV device performance for a fixed value of short circuit current density, J_{sc}=8.5 A/cm^2.

A dual junction TPV cell made from 343 meV and 450 meV bandgap materials modeled here will have nearly optimal performance for TPV power generation.[18] The 343 meV bottom junction receives illumination up to the 450 meV bandgap energy of the top junction. A well-defined spectrum of thermal emission gives an

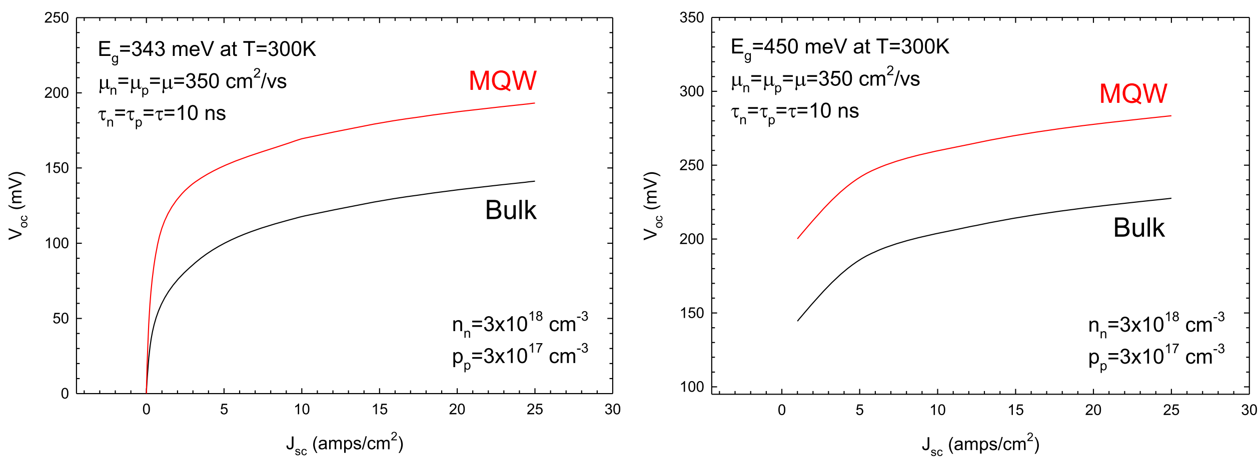

FIG. 2. Open circuit voltage vs. short circuit current at 300K for MQW and bulk IV-VI semiconductor materials with bandgaps of 343 meV and 450 meV.

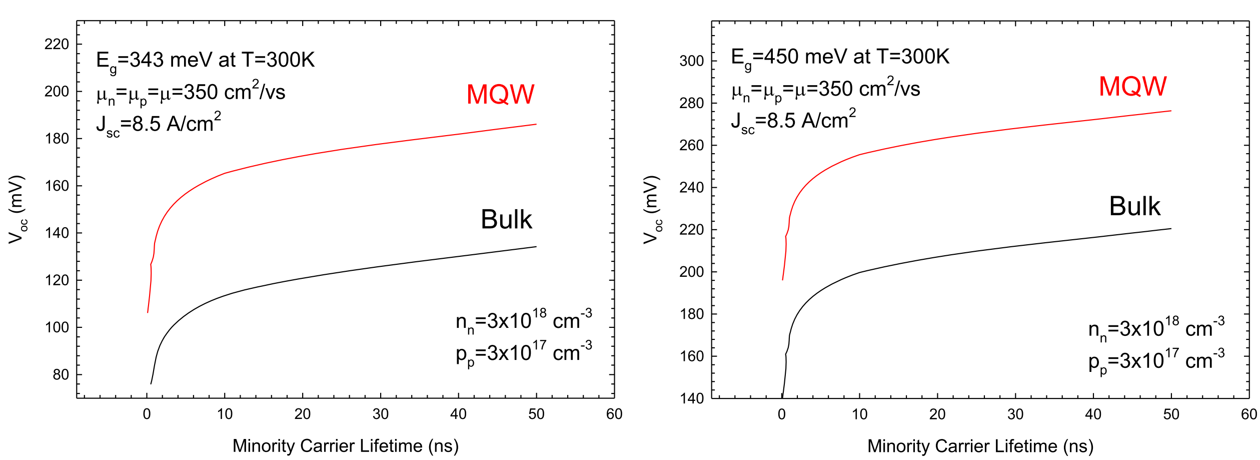

FIG. 3. Open circuit voltage vs. minority carrier lifetime at 300 K for MQW and bulk IV-VI semiconductor materials with bandgaps of 343 meV and 450 meV.

TABLE IV. Predicted TPV device performance parameters for four different narrow bandgap materials in a thermal generator with a 1215°C radiator (emissivity=0.9) with the device material at 300K and assuming a minority carrier lifetime of 10 ns.

Material (300K)	E_g	J_{sc} (A/cm^2)	V_{oc} (mV)	FF	P_{EL} (W/cm^2)
$Pb_{0.937}Sr_{0.063}Se$	450	8.5	199	0.64	1.08
$Pb_{0.976}Sr_{0.024}Se$	343	8.5	114	0.51	0.49
MQW (3.5 nm)	450	8.5	256	0.69	1.50
MQW (4.9 nm)	343	8.5	164	0.60	0.84

optical power density of 4.47 W/cm^2 for a 1215°C radiator with an emissivity of 0.9 for the bottom junction. A short circuit current density of 8.5 A/cm^2, the value used for plotting the data in Figure 3, is obtained for an average photon energy of 396 meV and an EQE of 75%. Figure 3 illustrates the key importance of a minority carrier lifetime. It shows that a IV-VI semiconductor device structure will need to have lifetimes greater than about 2 ns to achieve practical open circuit voltages.

Table IV shows short circuit current, open circuit voltage (assuming a 10 ns lifetime), FF, and electrical power density for the four different materials modeled here. FF values were determined using equations given in Ref. 19. Results show that a current-matched dual junction TPV cell made from MQW materials will have a total power density of 2.34 W/cm^2, 49% better than the 1.57 W/cm^2 generated by a dual junction cell made from bulk materials with the same bandgap energies. This improvement is a direct result of band structure engineering using quantum confinement effects. MQW materials remove L-valley degeneracy, which decreases thermally-generated charge carrier density and increases V_{oc}.

IV-VI semiconductor MQW materials are attractive for multi-junction TPV cell fabrication since materials with different bandgap energies can be made using just two alloys, $Pb_{0.80}Sr_{0.20}Se$ and $Pb_{0.81}Sn_{0.19}Se$. An MBE chamber with only six effusion cells, PbSe, SnSe, Sr, Se, and two dopant cells, can be used to grow structures for multi-junction TPV cells. Junctions with different bandgap energies are grown by specifying different well widths, which are controlled by timed shutter openings. Multi-junction TPV device optimization, especially the task of current matching, can be precisely and methodically conducted using this bandgap control technique. It is no more expensive to grow MQW materials in an MBE system than it is to grow bulk materials. The estimated cost for compound semiconductor device materials grown by MBE is in the range of \$1.00/cm^2 when scaled to high volume manufacturing.[20] Including the \$0.15/cm^2 cost for the silicon substrate, the cost for the IV-VI semiconductor MQW materials in an optimized dual junction TPV cell having a bottom junction bandgap of 343 meV will be less than \$0.50/W, about two orders of magnitude less expensive that currently available III-V semiconductor TPV materials.

An important question for determining whether or not a IV-VI semiconductor MQW materials technology is viable for TPV cell fabrication is related to epitaxial layer material quality. Lattice parameter mismatch between the thin layers of $Pb_{0.80}Sr_{0.20}Se$ (a_o = 0.615 nm) and $Pb_{0.81}Sn_{0.19}Se$ (a_o = 0.610 nm) in the MQW material will create 0.41% compressive and tensile strains, respectively, in each layer. The critical layer thickness at which this strain provides enough energy to create misfit dislocations is in the range of 50 monolayers (17.5 nm).[21] The MQW materials modeled here will therefore remain fully strained without creation of lattice mismatch defects. This strain, however, will slightly alter the well and barrier bandgaps through deformation potential effects.[22] In-plane compressive strain of the $Pb_{0.80}Sr_{0.20}Se$ barrier layer will increase its bandgap by about 7.5 meV, while in-plane tensile strain of the $Pb_{0.81}Sn_{0.19}Se$ well layer will decrease its bandgap by about 7.5 meV thus making well depths deeper by about 15 meV. This strain effect along with the effects of temperature and band non-parabolicity will be more fully evaluated in future papers.

The key issue for determining the technical viability of IV-VI semiconductor MQW materials for TPV cell fabrication is minority carrier lifetime. As Figure 3 clearly shows, lifetimes must be longer than a few nanoseconds for open circuit voltages to be large enough for practical TPV power generation. Crystalline defects that enhance electron-hole pair recombination are of particular concern, especially those that are due to surface chemistry effects. Limited work exists on the fabrication of IV-VI semiconductor optoelectronic devices designed for room temperature operation. Although there are experimental data supporting a 10 ns minority carrier lifetime in a PbSrS/PbS heterostructure on silicon,[11] they were obtained in an optically-pumped device without the electrical contacts that would be needed for a TPV device. It is therefore critical that the device processing procedures for forming ohmic contacts to the n-type and p-type sides of a TPV cell avoid formation of lifetime reducing defects. The next step for future work in this area is fabrication of metallized photovoltaic pn junction devices that employ IV-VI semiconductor epitaxial layers grown on silicon substrates. Demonstration of open circuit voltages greater than about 100 mV under thermal radiation illumination conditions will establish a significant proof-of-concept for the fabrication of low cost and high performance TPV cells for electrical power generation.

In summary, it has been shown that IV-VI semiconductor MQW materials can be designed to exploit L-valley degeneracy removal thereby reducing the effective densities of states for electrons and holes. This work represents a novel use of quantum wells. Quantum wells have been described for III-V semiconductor TPV devices where dark currents are reduced.[23] However, the benefit in this case was due to the barrier material having a larger bandgap and not a modification of the active material band structure as is possible with IV-VI semiconductors. The IV-VI semiconductor MQW materials described here, both of which have bandgap energies below 500 meV, are ideal candidates for development of multi-junction TPV cells. Such a development effort can lead to optimized TPV cells that achieve thermal engine efficiencies above 50%.[18] The materials technology described in this paper offers an attractive pathway for the development of commercially viable TPV-based electrical generators. These generators will be significantly less expensive to purchase and to operate than existing turbine or piston engine based generators.

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